The Washington Trails Scraper User manual

For Windows

By Calvin Bellamy

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Introduction to software

```
[INFO] ------------------------< com.wta:wta-scraper >-------------------------
[INFO] Building wta-scraper 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ wta-scraper ---
[INFO] skip non existing resourceDirectory C:\Users\[redacted]\Downloads\files\src\main\resources
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ wta-scraper ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- exec:3.6.3:java (default-cli) @ wta-scraper ---
=== WTA Hike Scraper ===
Scraping up to 500 hikes...
```

Figure 1. Initial scrape startup screen.

The Washington Trails Scraper is a custom program that take the 4214 trails in Washington and turns it into a formatted data list able to be used in programs or documents. Each trail is stored under the name of the trail, the data from the trail is stored under different elements: name, areaLocation, lengthMiles highestPointFeet, elevationGainFeet, tags, and url.

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Field	Type	Example
`name`	String	"Lake Serene"
`areaLocation`	String	"Index Area"
`lengthMiles`	double	8.0
`highestPointFeet`	double	5,200
`elevationGainFeet`	double	2,200
`tags`	List\<String\>	["Dogs allowed on leash", "Wildflowers/Meadows"]
`url`	String	"https://www.wta.org/go-hiking/hikes/lake-serene"

The software is able to scrape the data of any number of trials you need, this is done by entering a custom amount when running the program (default 60).

Installation

Starting: The software needed is provided along with this manual, download the two zip files (apache-maven-3.9.12-bin.zip and files.zip) and unzip them both in an accessible location,

1. Press the **Windows key** and search **"environment variables"**
2. Click the application named **"Edit the system environment variables"**

   a terminal should popup labeled System Properties.
3. At the bottom of the terminal click the **"Environment Variables..."** button under startup and recovery box (see figure 2)

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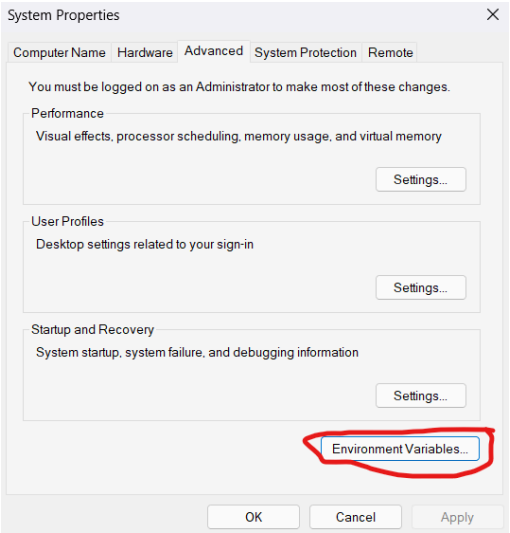

Figure 2

4. In the **"System variables"** section (bottom half), scroll down and click on **"Path"** → click **"Edit"** (See figure 3)

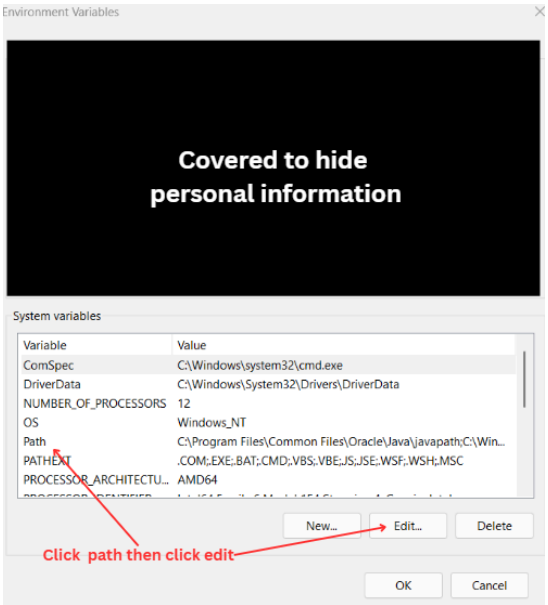

Figure 3

5. Go to where you stored your **apache-maven-3.9.12-bin file** and copy the address location from the address bar at the top of your files
   should look like this: C:\Users___\fileLocation\apache-maven-3.9.12-bin

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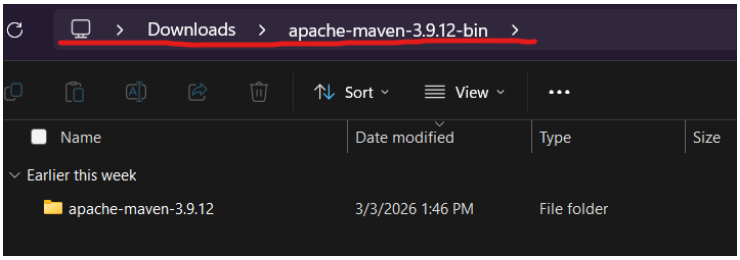

Figure 4

To verify the if the installation was successful open a new terminal by pressing the windows key and searching terminal, then in the terminal type "***mvn -version,***" it should print out something similar to Apache Maven 3.x.x, if it instead shows "mvn not recognized" then double check where you pasted your path and that your path matches the location of the zip.

Startup and Operation

To start the scraper:

1. Got to where you have your files.zip and extract it if you haven't already
2. Open the files file and in the address bar type powershell (See figure 5)

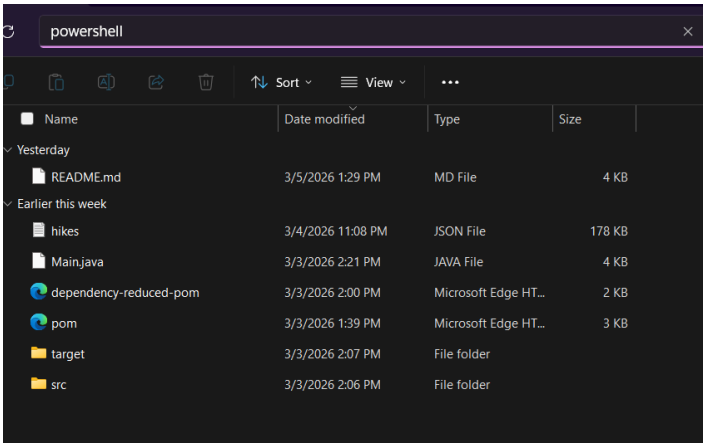

Figure 5

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This will open up a PowerShell Terminal directed to at your folder without commands

3. In the terminal type “mvn clean package” this makes Maven download the dependencies from the internet (this only happens the first time), Compile all the Java files, and Bundle everything into a single runnable jar file at *target/wta-scraper.jar.*

   Following this command the program will print text showing its operations, at the end it will either show BUILD SUCCESS or BUILD FAILURE, if the build success then your ready to start the next part, if it fails however check if you code is up to date with the current version of the program and or java.

To operate the program:

1. Either reopen powershell using previous steps from Startup or if you didn't close the previous powershell window keep it open.
2. Next in the powershell terminal type “***mvn compile exec:java "-Dexec.mainClass=com.wta.scraper.Main" "-Dexec.args=x’”***” where x is the amount of hikes you want.

   The program will now begin and will start scraping hikes from the Washington Trail Association website and store the data of the trails in “*hikes.json*”
   The runtime depends on the number of trails you are scraping. Each trail takes approximately 2.5 seconds, so the total time can be calculated with the equation: (Number of Trails × 2.5 seconds) + startup time (about 5 seconds) = total runtime.

Uses of Data

In *hikes.json* the data from the scraper is stored in a easy accessible format that can be used in a variety of Computer languages for analysis, you can filter it using the elements provided and their data, use it for averages, or even for your own personal library

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